

Ahsanullah University of Science & Technology
Department of Computer Science & Engineering
Course No: CSE 4107 Quiz #1 (Sec A) (Date: 21.05.18)
Semester: Spring 2018

Marks: 20**Time: 25 min**

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1. Which of the following are true and which are false? Explain your answers. **[4*2=8]**
 - a. $h(n) = 0$ is an admissible heuristic for the 8-puzzle.
 - b. A* is of no use in robotics because percepts, states, and actions are continuous.
 - c. Breadth-first search is complete even if zero step costs are allowed.
 - d. Assume that a rook can move on a chessboard any number of squares in a straight line, vertically or horizontally but cannot jump over other pieces. Manhattan distance is an admissible heuristic for the problem of moving the rook from square A to square B in the smallest number of moves.
 2. Consider a state space where the start state is number 1 and each state k has two successors: numbers $3k$ and $3k + 1$. **[3*4=12]**
 - a. Draw the portion of the state space for states 1 to 15.
 - b. Suppose the goal state is 16. List the order in which nodes will be visited for breadth first search, depth-limited search with limit 3, and iterative deepening search.
 - c. How well would bidirectional search work on this problem? What is the branching factor in each direction of the bidirectional search?

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1. Which of the following are true and which are false? Explain your answers. **[4*2=8]**
 - a. Depth-first search always expands at least as many nodes as A* search with an admissible heuristic.
 - b. $h(n) = 0$ is an admissible heuristic for the 8-puzzle.
 - c. A* is of no use in robotics because percepts, states, and actions are continuous.
 - d. Breadth-first search is complete even if zero step costs are allowed.
 2. Consider a state space where the start state is number 1 and each state k has two successors: numbers $2k$ and $2k + 1$. **[3*4=12]**
 - a. Draw the portion of the state space for states 1 to 15.
 - b. Suppose the goal state is 11. List the order in which nodes will be visited for breadth first search, depth-limited search with limit 3, and iterative deepening search.
 - c. How well would bidirectional search work on this problem? What is the branching factor in each direction of the bidirectional search?

Ahsanullah University of Science & Technology
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Course No: CSE 4107 **Quiz #1 (Sec B) (Date: 19.05.18)**
Semester: Spring 2018

Marks: 20**Time: 20 min**

1. Why IDS is better than BFS and DFS algorithms? Write the properties of Iterative Deepening Search (IDS).
2. Here is a weighted graph along with the heuristic function values of the nodes presented in tables.

Node	A	A	A	B	B	C	C	D	E
Neighbor	C	B	E	D	E	E	D	F	F
Cost	30	45	75	33	42	36	37	20	30

Node	A	B	C	D	E	F
H(Node)	65	58	62	36	30	0

Consider that A is the initial node and F is the goal node, and generate an optimal solution using A* Search.

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Course No: CSE 4107 **Quiz #1 (Sec B) (Date: 19.05.18)**
Semester: Spring 2018

Marks: 20**Time: 20 min**

1. Draw the state space for the 8-puzzle problem up to depth 2. The initial state is given below:

1	4	3
7		6
5	8	2

2. Here is a weighted graph along with the heuristic function values of the nodes presented in tables.

Node	A	A	A	B	B	C	C	D	E
Neighbor	C	B	E	D	E	E	D	F	F
Cost	35	40	70	33	42	36	37	30	29

Node	A	B	C	D	E	F
H(Node)	70	55	62	46	30	0

Consider that A is the initial node and F is the goal node, and generate an optimal solution using A* Search.

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Course No: CSE 4107 Quiz #2 (Sec A) (Date: 02.07.18)
Semester: Spring 2018

Marks: 20

Time: 25 min

1. What are the advantages of using Pattern databases? Describe Local Beam Search algorithm. [5+5]

2. Construct a pruned game tree using alpha-beta pruning. Take the sequence, [10]

[5, -2, -8, -13, -12, 3, -12, 3, -5]

of MINIMAX values for the nodes at the cutoff depth of 2 plies. Assume that branching factor is 3, **MAX** makes the first move, and nodes are generated from left to right.

Ahsanullah University of Science & Technology
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Course No: CSE 4107 Quiz #2 (Sec A) (Date: 02.07.18)
Semester: Spring 2018

Marks: 20

Time: 25 min

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1. How to generate admissible heuristics from relaxed problems? Explain with examples.
Describe Stochastic Hill climbing Search algorithm.

[5+5]

2. Construct a pruned game tree using alpha-beta pruning. Take the sequence,

[10]

[-7, 6, 9, -17, -6, -19, -12, -14]

of MINIMAX values for the nodes at the cutoff depth of 2 plies. Assume that branching factor is 3, **MAX** makes the first move, and nodes are generated from left to right.

Time: 20 min

1. Give the name of the algorithm that results from each of the following special cases: [1+1]
 - a. Local beam search with $k = 1$.
 - b. Genetic algorithm with population size $N = 1$.
2. Consider the problem of solving two 8-puzzles. Suppose we make the problem adversarial as follows: the two players take turns moving; a coin is flipped to determine the puzzle on which to make a move in that turn; and the winner is the first to solve one puzzle. Give an informal proof that someone will eventually win if both play perfectly. [3]
3. Construct a pruned game tree using alpha-beta pruning. Take the sequence, [10]
[**-10, -5, 15, 12, -8, -19, -19, 9**]
of MINIMAX values for the nodes at the cutoff depth of 3 plies. Assume that branching factor is 2, **MAX** makes the first move, and nodes are generated from left to right.

Time: 20 min

1. Give the name of the algorithm that results from each of the following special cases:
 - a. Local beam search with one initial state and no limit on the number of states retained.
 - b. Simulated annealing with $T = 0$ at all times (and omitting the termination test). [1+1]
2. Consider the problem of solving two 8-puzzles. Suppose we make the problem adversarial as follows: the two players take turns moving; a coin is flipped to determine the puzzle on which to make a move in that turn; and the winner is the first to solve one puzzle. Give an informal proof that someone will eventually win if both play perfectly. [3]
3. Construct a pruned game tree using alpha-beta pruning. Take the sequence, [10]
[20, -2, -19, -6, -6, 17, 18, 6]
of MINIMAX values for the nodes at the cutoff depth of 3 plies. Assume that branching factor is 2, **MIN** makes the first move, and nodes are generated from left to right.

Ahsanullah University of Science & Technology
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Course No: CSE 4107 Quiz #3 (Sec A/B/C) (Date: 31.07.18)
Semester: Spring 2018

Marks: 20**Time: 30 min**

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1. Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables or the equivalence rules. **[2X3=6]**
 - a. $(\text{Smoke} \Rightarrow \text{Fire}) \Rightarrow (\neg \text{Smoke} \Rightarrow \neg \text{Fire})$
 - b. $\text{Big} \vee \text{Dumb} \vee (\text{Big} \Rightarrow \text{Dumb})$
 2. Which of the following are correct? Explain. **[2X2=4]**
 - a) $A \Leftrightarrow B \models A \vee B$.
 - b) $((A \vee B) \wedge \neg(A \Rightarrow B))$ is satisfiable.
 3. Convert the following set of sentences to clausal form and use resolution to prove the sentence $\neg F$ from the clauses. **[10]**

S1: $\neg A$; S2: $\neg B \vee \neg C \vee A$; S3: B ; S4: $\neg D \vee \neg E \vee C$; S5: D ; S6: $E \vee F$

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Course No: CSE 4107 Quiz #3 (Sec A/B/C) (Date: 31.07.18)
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Marks: 20**Time: 25 min**

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1. Decide whether each of the following sentences is valid, unsatisfiable, or neither. Verify your decisions using truth tables or the equivalence rules. **[3X2=6]**
 - a) $\text{Smoke} \vee \text{Fire} \vee \neg \text{Fire}$
 - b) $(\text{Smoke} \Rightarrow \text{Fire}) \Rightarrow ((\text{Smoke} \wedge \text{Heat}) \Rightarrow \text{Fire})$
 2. Which of the following are correct? Explain. **[2X2=4]**
 - a) $(A \wedge B) \models (A \Leftrightarrow B)$.
 - b) $A \Leftrightarrow B \models \neg A \vee B$.
 3. Convert the following set of sentences to clausal form and use resolution to prove the sentence $\neg A \wedge \neg B$ from the clauses. **[10]**

S1: $A \Leftrightarrow (B \vee E)$; S2: $E \Rightarrow D$; S3: $C \wedge F \Rightarrow \neg B$; S4: $E \Rightarrow B$; S5: $B \Rightarrow F$; S6: $B \Rightarrow C$